

Part 1

Method

The goal for this lab is to verify buoyancy. Specifically, we are looking to verify that

$$m_{max} = \rho_{water} V_{object}$$

Where m_{max} includes the mass of a boat and any masses on/in it. We will be constructing tin foil boats and placing masses on them until it eventually sinks. Ideally, we can show that m_{max} has some fixed value for a given volume of an object, and identify that tipping point.

We will be using statistical analysis tools such as propagated error and T-score(s) to show the accuracy (or not) of our experiments.

We find the volume of our boat through displacement of the water and measuring this displacement amount in $\frac{mL}{m}$.

Data

We note the following measurement uncertainty:

1. Masses are ± 0.01 grams
2. G (constant) is $\pm 0.05 \frac{N}{kg}$
3. The volume of a boat is $\pm 0.01 m^3$
4. The density of water is $\pm 5 kg$
5. The graduated cylinder is just $1mL (1 \cdot 10^{-6} m^3)$
 - You'll need to propagate uncertainty. The needed partial derivatives are below.

$$\frac{\partial(\rho V)}{\partial \rho} = V \quad \frac{\partial(\rho V)}{\partial V} = \rho$$

We say that our error for pV is:

$$\sqrt{\left(\frac{dpV}{dp} \cdot dp^2\right)^2 + \left(\frac{dpV}{dV} \cdot dV^2\right)^2}$$

We identify the volume of our tupperware container to be

$$189mL = 0.000189m^3 \pm 1 \cdot 10^{-6} m^3.$$

We measure the original mass of the container to be $6.40 \pm 0.01 g$

Lab 6: Whatever floats your boat

Connor Leu (ccl2396), Abdon Morales (am226923), Chris Solis (CMS8862)

From the original equation, we expect the maximum mass that can float to be $1000 \frac{kg}{m^3} \cdot 0.000189m^3 = 0.189kg$. However, there will be error propagation on this estimated measurement from the above formula:

$$\sqrt{((0.000189)(5))^2 + (1000(10^{-6}))^2} \\ = 0.001376$$

So, we calculate that the maximum mass we can float to be $189g \pm 1.4g$.

We added mass up to 174g before it finally sank at 175g.

So, the total mass (including the boat) is $180.4 \pm 0.01 g$

Computing a T-score for this compared to the ideal computed value:

$$\frac{|0.189 - 0.1804|}{\sqrt{0.001376^2 + 0.00001^2}} \\ = 6.249835$$

(bad)

Conclusion

We didn't meet the computed maximum mass, but we believe this to be due to a few factors. First, we can see from the T score of 6 that we are wildly distinguishable from the computed maximum mass. We identify the following primary error sources:

1. Our measurement capabilities for the volume of the boat being very simplistic
2. Unbalanced placement of the masses

In terms of absolute numbers, being ~9g off doesn't look too bad, but the T score says otherwise. We strongly believe that the balance of the weights played a big part in sinking earlier than we expected, as these tupperware boats are very unstable and prone to tipping.

Future iterations of the experiment should be measured more precisely to start with (see Part 2).

Our results line up with a fellow group.

Part 2

Method

Due to our distinguishable T-score result, we are redoing the experiment. We will change how we identify volumes (using mass/density calculations and not displacement measurements). We also attempt to center the masses in such a way that tipping is minimized. This time, we will try to use lots of very small masses to balance the load better.

Data

Our second measurement of the boat mass is $6.32g \pm 0.01g$

We find the volume of the boat this iteration to be $174\text{mL} = 0.000174\text{m}^3 \pm 1\text{e-}6\text{ m}^3$.

All other uncertainties remain the same.

We expect the maximum mass to be $1000 \frac{\text{kg}}{\text{m}^3} \cdot 0.000174\text{m}^3 = 0.174\text{kg} = 174g$

The error for this will be computed the same, and comes out to be $\pm 0.001325g$.

Adding masses to the boat, we sink at a combined mass of $180g$.

We T-score for this:

$$\frac{|0.174 - 0.18|}{\sqrt{0.001325^2 + \left(\frac{0.01}{1000}\right)^2}} = 4.528173$$

Conclusion

Although we did not meet the computed maximum mass of the boat, we successfully refined our experiment to improve precision and minimize errors, as demonstrated by lowering the t-score from 6.249 to 4.52. Despite this improvement, our data is still distinguishable according to the new t-score conducted from the newly-adjusted experiment; the possible reasons for our t-score to still be distinguishable is the possible lack of more precise weights (such as 1 gram or 10 gram weights) and the fact that the “boats” we were using are, for lack of a better term, horrible. We predict that in future experiments, if we could create a “boat” that had a much more stable base, that we could come much closer to our computed buoyancy value.